

Turing Way-Aligned Readiness Snapshot

This is a Turing Way-aligned readiness snapshot, not an official Turing Way certification. The Turing Way is a set of public best-practice guidelines, not a certifying authority. No institution or program has endorsed this repository as "certified." Scores below reflect only what is directly observable in the reviewed repository at the reviewed commit.

- **Repository:** [stjude-biohackathon/KIDS25-Team4](#)
- **Reviewed ref:** `main` @ commit [a8cec52](#)
- **Review date:** 2026-09-18
- **Scope:** Full public repository tree (root files, `UI Files/`, `igv/`, `nf_integrated/`, `api_server.py`, `run.sh`, `task/planning docs`). No sensitive/private data reviewed.
- **Confidence:** Medium. Evidence is based on direct inspection of the file tree, README/task docs, LICENSE, environment files, git history/authors, and code entry points. Deep line-by-line review of every bundled third-party script (e.g. ANNOVAR Perl scripts, Java jars) was not performed.

Score Summary

#	Criterion	Score (0–10)
1	Project purpose and scope	6
2	Version control and provenance	5
3	Open collaboration	2
4	Reproducible environments	4
5	Data and workflow provenance	3
6	Testing and validation	1
7	Automation and continuous integration	0
8	Documentation and usability	6
9	Licensing, attribution, and responsible reuse	5
10	Ethics, accessibility, and sustainability	1

#	Criterion	Score (0–10)
Total		33 / 100

Criterion Detail

1. Project purpose and scope — 6/10

Evidence: README.md states the project goal ("Build and modernize the pipeline for usability and reproducibility"), names the two modules (WGS, MitoEdit), and includes a task list and a Mermaid workflow diagram. **Gap:** No explicit statement of intended users, target audience, or known limitations/caveats of the pipeline. **Repository fact:** README.md describes the WGS-MTE pipeline goal and module breakdown — <https://github.com/stjude-biohackathon/KIDS25-Team4/blob/a8cec5270af8534953410c703d844cce5f50959a/README.md> **Turing Way guidance:** [Guide for Project Design](#)

2. Version control and provenance — 5/10

Evidence: 30 commits on main with descriptive, categorized messages (e.g. "Update [shell]:", "New [YAML, shell]:") from multiple named contributors, tracked in a standard GitHub repo. **Gap:** No tags or releases exist to mark reproducible snapshots; some commits bundle large third-party binary/annotation files together with pipeline changes, reducing change traceability. **Repository fact:** git log shows 30 commits with descriptive messages, no tags found — <https://github.com/stjude-biohackathon/KIDS25-Team4/commit/a8cec5270af8534953410c703d844cce5f50959a> **Turing Way guidance:** [Version Control](#)

3. Open collaboration — 2/10

Evidence: The repository is public and accepts GitHub's default issue/PR mechanisms; multiple named contributors appear in the git history. **Gap:** No CONTRIBUTING.md, CODE_OF_CONDUCT.md, issue templates, PR templates, or .github/ directory exist. There is no documented review process or maintainer list. **Repository fact:** No .github directory, CONTRIBUTING, or CODE_OF_CONDUCT files exist in the tree — <https://github.com/stjude-biohackathon/KIDS25-Team4/tree/a8cec5270af8534953410c703d844cce5f50959a> **Turing Way guidance:** [Handling Technical Issues in a New Community](#)

4. Reproducible environments — 4/10

Evidence: Per-module Conda environment files exist (igv/ends.yml, nf_integrated/nf_mte/mte.yml, nf_integrated/

Mito_WGS_Nextflow/conda_environment/nf_mitvar.yml ,
annot_python2.yml); README documents specific HPC module
versions used (e.g. samtools/1.12 , bwa/0.7.17). **Gap:** No
consolidated root-level environment/requirements file; no Docker/Singularity
container yet (listed as an open TODO in README/TASKS.md);
dependency pinning is inconsistent across the three separate .yml files.
Repository fact: Conda .yml files exist under igv/ and
nf_integrated/ subfolders — [https://github.com/stjude-biohackathon/
KIDS25-Team4/blob/a8cec5270af8534953410c703d844cce5f50959a/
nf_integrated/nf_mte/mte.yml](https://github.com/stjude-biohackathon/KIDS25-Team4/blob/a8cec5270af8534953410c703d844cce5f50959a/nf_integrated/nf_mte/mte.yml) **Turing Way guidance:** [Open Source](#)

5. Data and workflow provenance — 3/10

Evidence: Nextflow (Mitochondra_Variant_Call.nf) and shell
scripts encode workflow steps in code; a sample output fixture
(nf_integrated/nf_mte/test_WGS_output.tab) exists. **Gap:** No
standalone data dictionary/schema documentation, no described data
sources/provenance for reference/annotation files, and no explicit output-
format documentation outside inline code/comments. **Repository fact:**
nf_integrated/nf_mte/test_WGS_output.tab is a sample output
file without accompanying schema docs — [https://github.com/stjude-
biohackathon/KIDS25-Team4/blob/
a8cec5270af8534953410c703d844cce5f50959a/nf_integrated/nf_mte/
test_WGS_output.tab](https://github.com/stjude-biohackathon/KIDS25-Team4/blob/a8cec5270af8534953410c703d844cce5f50959a/nf_integrated/nf_mte/test_WGS_output.tab) **Turing Way guidance:** [Research Data Management Checklist](#)

6. Testing and validation — 1/10

Evidence: README/TASKS.md list a "Validation" task (compare Nextflow
outputs vs. legacy pipeline) as a planned, unchecked item. **Gap:** No
automated test suite, test runner, or CI-triggered checks exist. The only
filename containing "test" (test_WGS_output.tab) is a static data
fixture, not an executable test. **Repository fact:** Repository tree contains
only one file matching "test" (a data fixture), no pytest/unittest suite —
[https://github.com/stjude-biohackathon/KIDS25-Team4/tree/
a8cec5270af8534953410c703d844cce5f50959a/nf_integrated/nf_mte](https://github.com/stjude-biohackathon/KIDS25-Team4/tree/a8cec5270af8534953410c703d844cce5f50959a/nf_integrated/nf_mte)
Turing Way guidance: [Testing Guidance](#)

7. Automation and continuous integration — 0/10

Evidence: None found. **Gap:** No .github/workflows/ directory or any
other CI configuration exists; no automated linting, testing, or packaging
checks run on commits or pull requests. **Repository fact:** No .github/
workflows directory exists in the repository — [https://github.com/stjude-
biohackathon/KIDS25-Team4/tree/
a8cec5270af8534953410c703d844cce5f50959a](https://github.com/stjude-biohackathon/KIDS25-Team4/tree/a8cec5270af8534953410c703d844cce5f50959a) **Turing Way guidance:**
[Continuous Integration Practices](#)

8. Documentation and usability — 6/10

Evidence: Root `README.md` is well-structured with goals, a checklist-style task list, a clarified outline, and a Mermaid workflow diagram;

`api_server.py` opens with a module docstring describing its purpose;

`igv/README.md` and `UI Files/readme.MD` provide module-level

docs. **Gap:** No unified setup/onboarding guide spanning all modules; several core capabilities (UI, containerization, validation) remain unchecked TODOs in the README itself, and code-level docstrings are sparse outside

`api_server.py`. **Repository fact:** `api_server.py` opens with a module docstring describing the API server purpose — https://github.com/stjude-biohackathon/KIDS25-Team4/blob/a8cec5270af8534953410c703d844cce5f50959a/api_server.py

Turing Way guidance: [Documenting your Project](#)

9. Licensing, attribution, and responsible reuse — 5/10

Evidence: A permissive MIT `LICENSE` file exists at the repository root, correctly identifying the copyright holder (St. Jude Children's Research Hospital BioHackathon, 2025); the initial commit message credits the original Nextflow pipeline authors. **Gap:** No `CITATION.cff` or structured citation file exists; the repository bundles third-party tools (ANNOVAR Perl scripts, `bambino-1.0.jar`, `picard.jar`, `mysql-connector-java`) without accompanying attribution/license files for those components.

Repository fact: `LICENSE` file at repo root grants MIT license to St. Jude Children's Research Hospital BioHackathon (2025) — <https://github.com/stjude-biohackathon/KIDS25-Team4/blob/a8cec5270af8534953410c703d844cce5f50959a/LICENSE>

Turing Way guidance: [Licensing](#)

10. Ethics, accessibility, and sustainability — 1/10

Evidence: None found addressing ethics, data privacy, accessibility, or long-term maintenance/archival planning, despite the pipeline processing genomic sequencing data. **Gap:** No ethics/data-governance statement, no accessibility considerations, and no stated maintenance or archival plan beyond an open task list.

Repository fact: No ETHICS, ACCESSIBILITY, or MAINTENANCE documentation files found in the repository tree — <https://github.com/stjude-biohackathon/KIDS25-Team4/tree/a8cec5270af8534953410c703d844cce5f50959a>

Turing Way guidance: [Accessibility](#)

Recommendations (priority order)

1. Add a `.github/workflows/` CI pipeline (lint + a smoke test) — highest-impact gap (Criteria 6 & 7 currently score 0–1).

2. Add `CONTRIBUTING.md` and `CODE_OF_CONDUCT.md` to enable structured open collaboration (Criterion 3).
3. Consolidate environment files into one pinned, documented environment (or the already-planned Docker/Singularity container) at the repo root (Criterion 4).
4. Add a `CITATION.cff` and attribution notes for bundled third-party tools (Criterion 9).
5. Add a short ETHICS/DATA-GOVERNANCE and maintenance/archival note given the genomic data domain (Criterion 10).
6. Document data schemas/provenance for pipeline inputs and outputs (Criterion 5).

Limitations of this snapshot

- This is an automated, evidence-based snapshot generated with assistance from Turing Way RAG evidence retrieval; it is **not** a peer review, security audit, or institutional certification.
- Only the public repository tree at the pinned commit was inspected; private CI/organization-level policies (if any) outside this repo are not visible and were not scored.
- Scores are based on presence/absence of observable artifacts, not on runtime validation of the pipeline itself.

Total score: 33 / 100